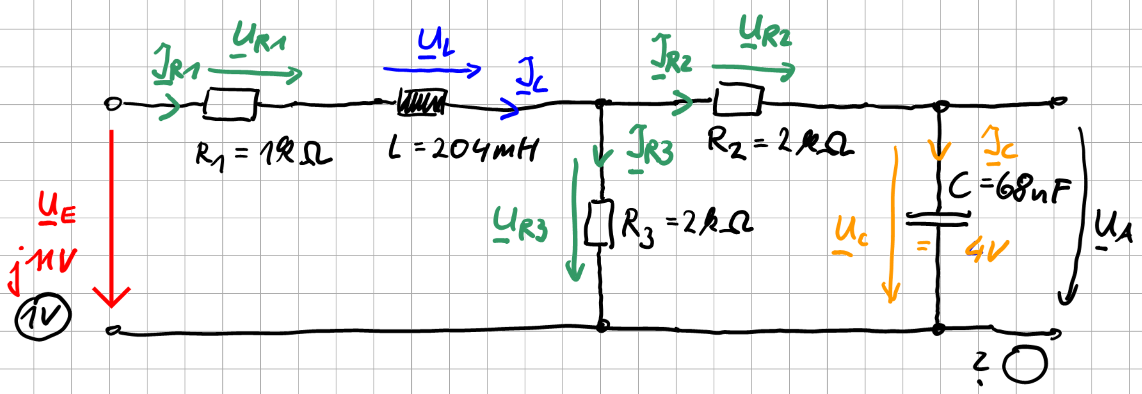
ZE11



$\omega = 7353 \frac{1}{s}$

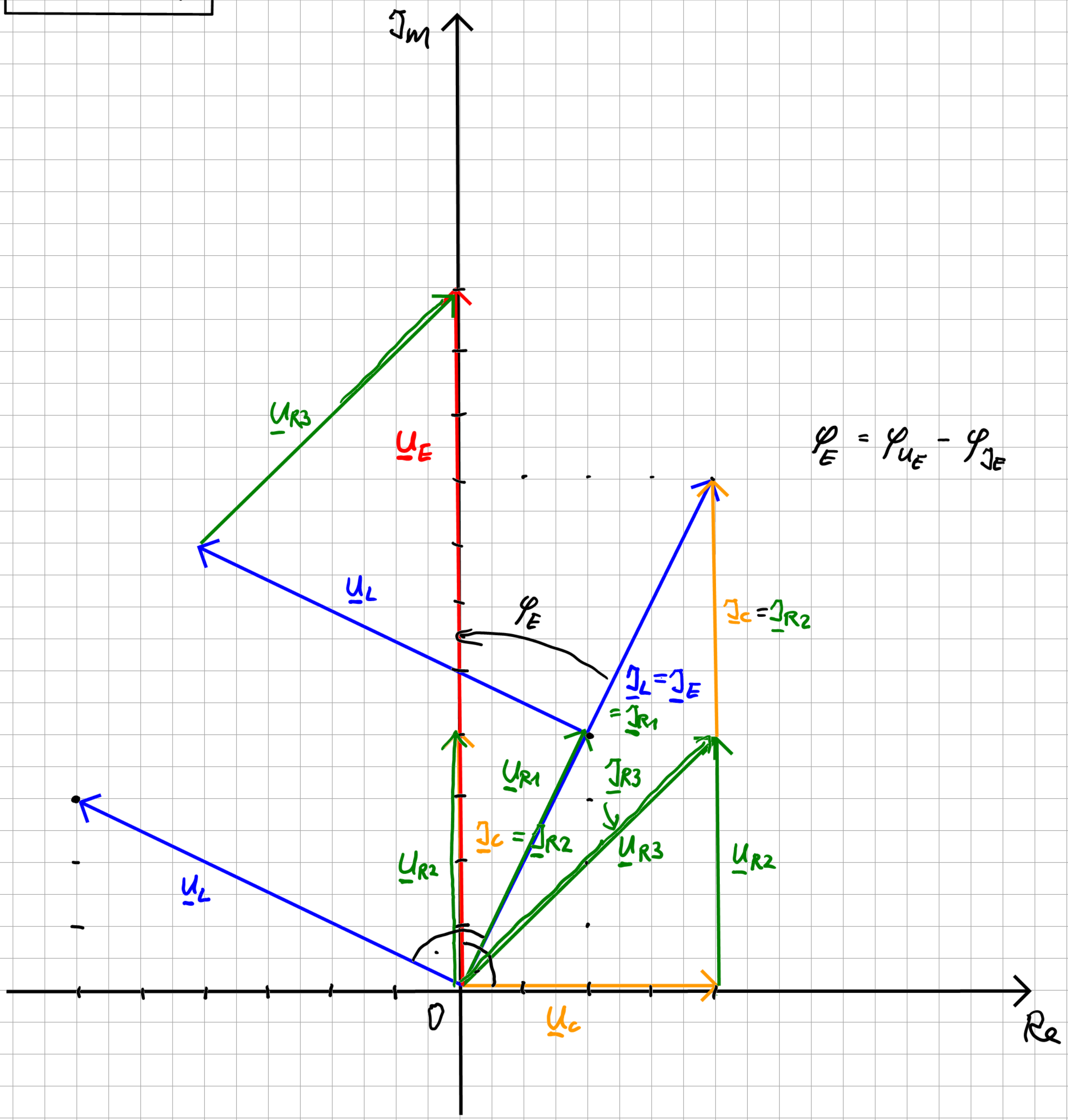
rechts

$|\underline{U}_A| \hat{=} 4\,cm$

$|\underline{I}_C| \hat{=} 4\,cm$

Freie Wahl Maßstab

$1V \hat{=} 1cm$
$1mA \hat{=} 2cm$



$\varphi_E = \varphi_{U_E} - \varphi_{I_E}$

$\underline{U}_A = \underline{U}_c = 4V + j0V$ (rechts)

$\underline{I}_c = \dfrac{\underline{U}_c}{\underline{Z}_c}$

$$\underline{Z}_c = \frac{1}{j\omega C} = \frac{1}{j\,7353\,\frac{1}{s}\cdot 68nF} = \frac{1}{j}\,2000\,\frac{1}{\frac{As}{V}\,\frac{1}{s}}$$

$$= -j\,2000\,\Omega = -j\,2k\Omega$$

$\underline{I}_c = \dfrac{4V}{-j\,2k\Omega} = +j\,2\,mA$ ($\hat{=}$ 4 cm)

$\Rightarrow$ Maßstab $1mA \,\hat{=}\, 2\,cm$

$\underline{I}_{R2} = \underline{I}_c = j2mA$

$\underline{U}_{R2} = R_2 \cdot \underline{I}_{R2}$

$= 2k\Omega \cdot j2mA = j\,4V$

$\underline{U}_{R3} = \underline{U}_{R2} + \underline{U}_c$

$= j4V + 4V = 4V + j4V$

$\underline{I}_{R3} = \dfrac{\underline{U}_{R3}}{R_3} = \dfrac{4V + j4V}{2k\Omega} = \underset{\text{4 cm rechts}}{2mA} + \underset{\text{4 cm oben}}{j2mA}$

$\underline{I}_{R1} = \underline{I}_L = \underline{I}_{R2} + \underline{I}_{R3}$

$= j2mA + 2mA + j2mA$

$= 2mA + j4mA$

$\underline{U}_L = \underbrace{j\omega L}_{\underline{Z}_L} \cdot \underline{I}_L$

$$\underline{Z}_L = j\omega L = j\,7353\,\frac{1}{s}\cdot 204\,mH = j\,1500\,\Omega$$

$\underline{U}_L = j\,1500\,\Omega \cdot (2mA + j\,4mA)$

$= j\,3V + j^2\,6V$

$= -6V + j3V$

$$\underline{U}_{R1} = R_1 \cdot \underline{I}_{R1}$$
$$= 1\,k\Omega \cdot (2mA + j\,4mA)$$
$$= 2V + j\,4V$$

$$\underline{U}_E = \underline{U}_{R1} + \underline{U}_L + \underline{U}_{R3}$$
$$= 2V + j\,4V - 6V + j\,3V + 4V + j\,4V$$
$$= j\,11\,V$$

b) Ähnlichkeit

$\underline{U}_A = 4\,V \quad \Rightarrow \quad \underline{U}_E = j\,11\,V$

$\underline{U}_A = x \quad \Leftarrow \quad \underline{U}_E = 1\,V$

$$\frac{4V}{x} \mathrel{\widehat{=}} \frac{j\,11\,V}{1V}$$

$$\Rightarrow x = \frac{4V \cdot 1V}{j\,11\,V} = -j\,\frac{4}{11}\,V$$

c) $$\varphi_E = \varphi_{U_E} - \varphi_{I_E}$$
$$= \measuredangle\left(\frac{\underline{U}_E}{\underline{I}_E}\right)$$

wohin

woher

$$\varphi_{U_E} = \measuredangle(\underline{U}_E) = +90°$$
$$\varphi_{I_E} = \measuredangle(\underline{I}_E) = \measuredangle(2mA + j\,4mA)$$
$$= \arctan\left(\frac{4mA}{2mA}\right) = 63{,}4°$$

$$\varphi_E = 90° - 63{,}4° = 26{,}6°$$

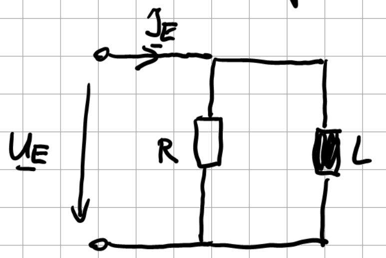
d) Wir suchen Ersatzschaltung bezüglich Eingang:

$$\underline{U}_E = j\,11\,V$$

$$\underline{I}_E = 2mA + j\,4mA$$

Parallelschaltung:

$\underline{Y}_E \Rightarrow$



$$\underline{Y}_E = \frac{\underline{I}_E}{\underline{U}_E} = \frac{1}{R} + \frac{1}{j\omega L}$$

$$\frac{2mA + j\,4mA}{j\,11\,V} = -j\,\frac{2}{11}\,mS + \frac{4}{11}\,mS \overset{!}{=} \frac{1}{R} + \frac{1}{j\omega L}$$

$$\Rightarrow \frac{1}{R} = \frac{4}{11}\,mS$$

$$R = \frac{11}{4}\,k\Omega = 2{,}75\,k\Omega$$

$$\frac{1}{j\omega L} = -j\,\frac{2}{11}\,mS$$

$$\cancel{-j}\,\frac{1}{\omega L} = \cancel{-j}\,\frac{2}{11}\,mS$$

$$\omega L = \frac{11}{2\,mS}$$

$$L = \frac{1}{\omega}\;\frac{11}{2}\,k\Omega$$

$$= \frac{1}{7353\,\frac{1}{s}}\;\frac{11}{2}\,k\,\frac{V}{A} \qquad \frac{Vs}{A} = H$$

$$= 750 \cdot 10^{-6}\,mH$$

$$= 0{,}75\ \ H$$

Ersatzschaltung in Reihe:

$$\underline{Z}_E = \frac{\underline{U}_E}{\underline{I}_E} = \ldots = Re + j\,Im$$

$$= \frac{j\,11\,V}{2\,mA + j\,4\,mA}$$

$$= \frac{j\,11\,V}{\sqrt{(2\,mA)^2 + (4\,mA)^2} \cdot e^{j\arctan\left(\frac{4\,mA}{2\,mA}\right)}}$$

$$= \frac{j\,11\,V}{4{,}47\,mA \cdot e^{j\,63{,}4°}}$$

$$= \frac{11\,V}{4{,}47\,mA}\, e^{j\,90°} \cdot e^{-j\,63{,}4°}$$

$$= 2460\,\Omega \cdot e^{j\,26{,}6°}$$

$$= 2460\,\Omega \cdot \cos(26{,}6°) + j\,2460\,\Omega \cdot \sin(26{,}6°)$$

$$= \underbrace{2200\,\Omega}_{R} + \underbrace{j\,1100\,\Omega}_{j\omega L}$$

$$= R + j\omega L$$

$$\omega L = 1100\,\Omega$$

$$L = \frac{1100\,\Omega}{7353\,\frac{1}{s}} = 0{,}15\,H$$